

A Case of Acute Coronary Syndrome and Cardiogenic Shock Managed with Stenting Across Large Serial LAD Coronary Artery Aneurysms

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INTRODUCTION

- Coronary artery aneurysms (CAA) are rare, heterogeneous lesions that are classified by location, size, morphology (saccular, fusiform, giant), and associated pathophysiology
- ACS + CAA represents a challenging clinical scenario with management guided largely by expert consensus, without robust outcomes data to support different modalities of intervention

CASE PRESENTATION

- 42M, no PMH; 30-min acute substernal chest pain
- ECG: anterior STEMI (STE V1–V4) → emergent cath lab activation

Emergent Angiography

- 100% proximal LAD thrombotic occlusion treated with POBA, revealing serial mid-distal LAD CAAs with heavy thrombus burden; further PCI deferred
- IABP placed; admitted to CICU on cangrelor; expedited Heart Team evaluation

CICU Course

- TTE: EF 47%, global hypokinesis (stunned myocardium)
- CTA head-to-toe: no extra-coronary aneurysmal disease, no aortopathy
- Interventional Plan: CABG x2 (LIMA-LAD, radial-OM1) + LAD ligation + pericardial patch occlusion of LM

Intraoperative Complication

- Toward conclusion of the case, severe acute systolic dysfunction (EF <10%) developed → placed back on full CPB support, LM patch removed, LAD ligation reversed → transferred urgently to cath on pressor and IABP support

Salvage PCI

- IVUS-guided DES: 3.0×48mm + 3.0×32mm overlapping Synergy DES, ostial–mid LAD achieved complete CAA sac exclusion; 0% residual stenosis

Outcomes

- Weaned from pressors POD 4; discharged home HD 18
- 6-month coronary CTA demonstrated stents patent with shrinkage of excluded thrombus

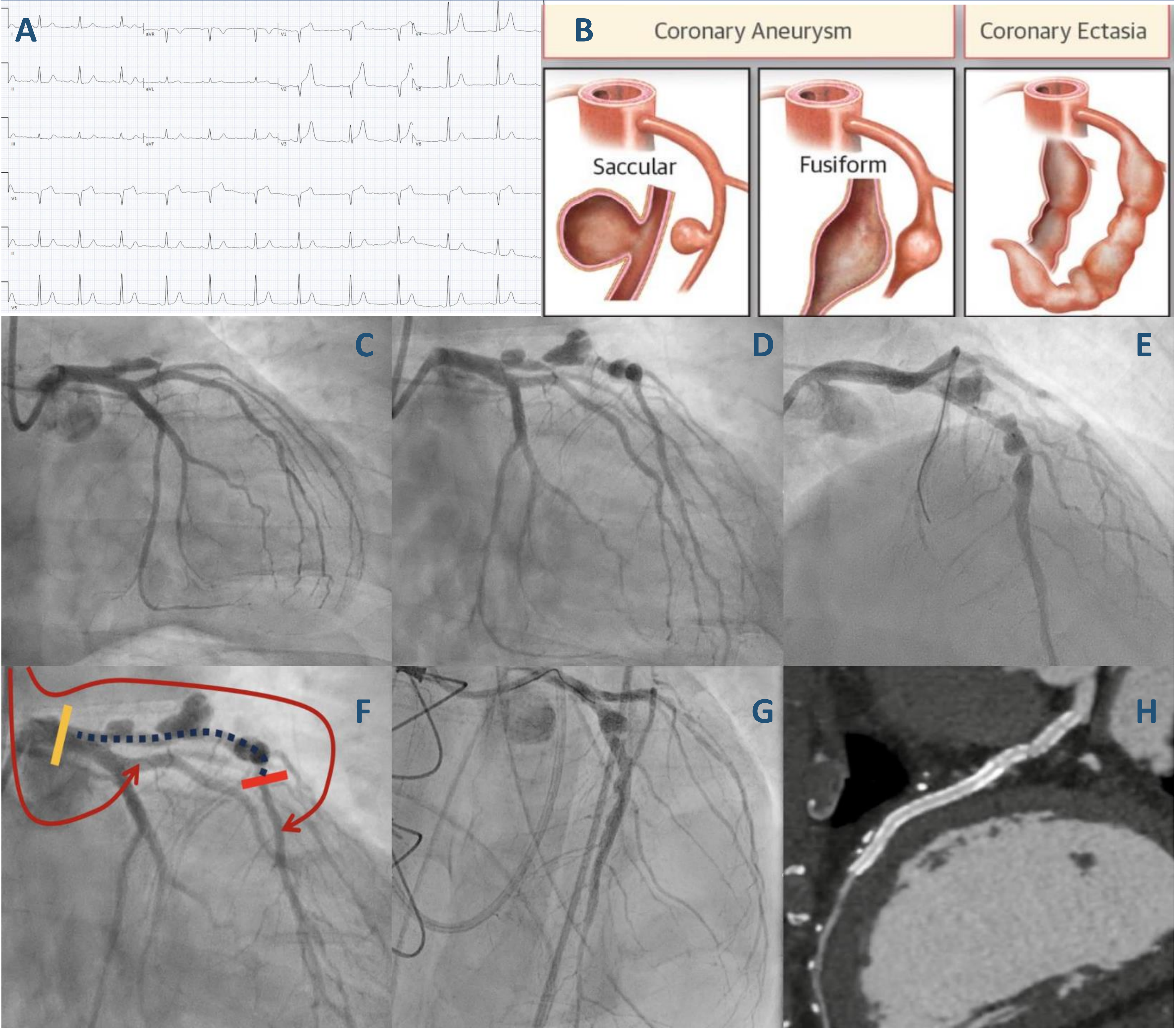
DISCUSSION

- Management of CAAs, aside from immediate balloon reperfusion in the ACS setting, is driven largely by multidisciplinary expert consensus.
- There is little high-quality evidence to inform superiority of either surgical (e.g., ligation or resection and CABG) or percutaneous intervention; similarly little evidence to guide different percutaneous options, i.e., DES-only, covered-stent, and/or coil-embolization behind the DES or covered stent.

REFERENCES

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FIGURES



A. Presenting ECG, demonstrating anterior STEMI. B. Different morphologies of CAA (extracted from Kawsara et al., JACC 2018). C. Initial angiogram of left system during emergent coronary angiography with complete LAD occlusion. D & E. Angiograms post-POBA demonstrating serial CAA in proximal LAD. F. Planned surgical intervention (red arrows CABG, yellow bar LM patch, red bar LAD ligation). G. Final angiogram after salvage PCI to LAD with overlapping DES. H. Six month follow up coronary CTA demonstrating patent stent with fully excluded thrombus.